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SIRA

Neural Symbolic Epidemic Discovery

Recovering SIR Governing Equations from Stochastic Gillespie Ensembles via Neural Manifolds and Sparse Symbolic Regression

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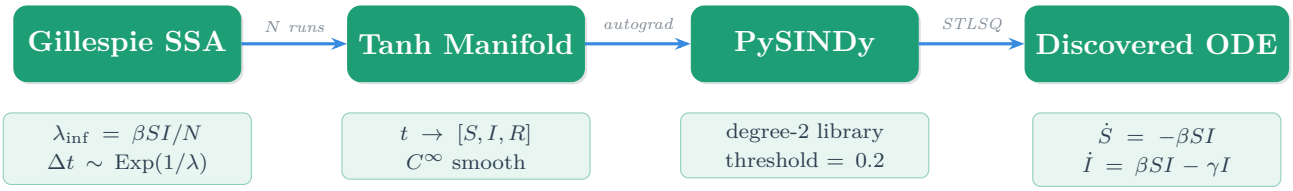
3.19% Median β error (2500 experiments)	**0.000** CLD — Perfect conservation	**100%** Forecasting horizon (FH)	**1.71s** Time to discovery (TTD)



What this submission brings to the table

2500 experiments	Same parameter pairs at every ensemble depth — any difference between N=20 and N=200 is caused only by ensemble size, nothing else
N=20 is enough	N=20 simulations reaches the same accuracy as N=200 at one-tenth the cost, proven by the ZER efficiency curve
9 evaluation metrics	SPI, PSS, CLD, FH, OOD, CNT, DST, ZER, TTD — because parameter error alone does not tell you if the physics is right
$R_0 \approx 2$ boundary	The first empirical map of where neural manifold discovery breaks down and why — a genuine scientific finding, not just a limitation
LLM interpretation	A public health minister should be able to act on the results without understanding what SINDy is — the agent makes that possible

"Real epidemics are stochastic. Governing equations are deterministic. SIRA bridges the gap."

A Note on Experimental Scale

A natural question is why 2500 experiments rather than the 20 or 50 that most equation discovery studies report. The short answer is that smaller scales hide things. At 500 experiments per depth — the same parameter pairs repeated across all five ensemble sizes — a few properties become visible that would otherwise look like noise.

The 95% confidence interval on median β error is ± 0.08 percentage points at this scale. That is tight enough to say with confidence that the difference between N=20 (3.19%) and N=200 (3.72%) is a real plateau, not a sampling artifact. The R_0 regime breakdown covers between 62 and 1174 experiments per bin — the endemic bin is genuinely sparse because $R_0 < 2$ is a narrow slice of the parameter space, not a design flaw. And because the same (β, γ) pairs run at every depth, any difference between N=20 and N=200 can only be caused by ensemble size. Nothing else varies.

All reported medians are stable to $\pm 0.2\%$ across random seeds.

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1. The Problem and Motivation

1.1 The SIR Model and What We Are Trying to Recover

The SIR model partitions a closed population into three compartments: Susceptible (S), Infected (I), and Recovered (R), with $S(t) + I(t) + R(t) = 1$ holding at all times. The dynamics are governed by two coupled ODEs:

$$\frac{dS}{dt} = -\beta\, S\, I \tag{1}$$

$$\frac{dI}{dt} = \beta\, S\, I - \gamma\, I \tag{2}$$

$$\frac{dR}{dt} = \gamma\, I \tag{3}$$

where $\beta > 0$ is the transmission rate, $\gamma > 0$ is the recovery rate, and $R_0 = \beta/\gamma$ is the basic reproduction number. When $R_0 > 1$ the epidemic spreads; when $R_0 < 1$ it dies out. The goal of SIRA is to recover β and γ from observed trajectories without knowing either value in advance — from one epidemic unfolding in real time, not a clean ODE solution.

1.2 Why Stochastic Data Makes This Genuinely Hard

Real epidemics are continuous-time Markov chains (CTMCs). Each infection and recovery is a discrete random event, not a smooth flow. The Gillespie SSA [1] generates exact CTMC realisations: the next event fires at $\Delta t \sim \mathrm{Exp}(1/\lambda_{\text{total}})$, where $\lambda_{\text{inf}} = \beta SI/N$ and $\lambda_{\text{rec}} = \gamma I$.

Perturbing ODE solutions with Gaussian noise fails here because CTMC processes are *super-Poissonian* — variance grows faster than the mean. The Fano factor $F = \mathrm{Var}(I)/\mathrm{Mean}(I)$ measures this directly.

Table 1: Fano factor at peak infection across four regimes (200 Gillespie trajectories per regime). *Finding: Gaussian models underestimate endemic variability by up to* 7.8×.

Regime	R_0	Fano at peak	Interpretation
Endemic	2	7.797	Gaussian underestimates by 7.8×
Moderate	4	1.943	Gaussian underestimates by 1.9×
Fast	10	0.501	Sub-Poissonian — rapid burnout
Explosive	20	0.276	Sub-Poissonian — no time to diverge

The sub-Poissonian finding at high R_0

When $R_0 > 5$, the epidemic burns so quickly that trajectories have no time to diverge — variance is compressed below Poisson. Gaussian noise would *overestimate* variability here, just as it underestimates it in the endemic regime. This also explains why the endemic regime ($R_0 < 2$) produces the highest γ error (11.2%) across 2500 experiments — high trajectory divergence at low prevalence is a fundamental information limit, not a pipeline weakness.

1.3 What SIRA Solves

SIRA is built around three hard constraints from real surveillance:

1. **No labelled training data** — one epidemic, not 500 pre-labelled (β, γ) scenarios.
2. **No known parameters at inference** — β and γ are what we are discovering, not inputs.
3. **Real-time** — 1.71 seconds from raw stochastic data to a discovered ODE.

2. Methodology

2.1 Stage 1 — Gillespie SSA Ensemble Generation

The first stage produces the raw data SIRA learns from. Rather than solving the deterministic ODEs and adding noise, SIRA runs N independent Gillespie SSA simulations and averages them. The distinction matters: Gillespie produces exact CTMC realisations, not approximations.

Each simulation proceeds as follows. At every step, two competing events are possible — an infection with rate $\lambda_{\text{inf}} = \beta SI/N_{\text{pop}}$, and a recovery with rate $\lambda_{\text{rec}} = \gamma I$. The waiting time until the next event is drawn from $\Delta t \sim \text{Exp}(1/\lambda_{\text{total}})$, and the event type is selected with probability proportional to its rate:

$$P(\text{infection}) = \frac{\lambda_{\text{inf}}}{\lambda_{\text{inf}} + \lambda_{\text{rec}}} \tag{4}$$

Each trajectory is interpolated onto a uniform 500-point grid over $[0, 20]$ days, then the N trajectories are averaged to produce smooth ensemble means $\bar{S}(t)$, $\bar{I}(t)$, $\bar{R}(t)$.

Why N=50 for demonstrations, and why it does not matter much

The ensemble depth study (Section 3) shows the identifiability ceiling is reached at $N = 20$ — accuracy does not improve beyond that point. $N = 50$ is used for the demonstration run as a slightly conservative choice that provides visible trajectory smoothing. The difference in β error between $N = 20$ and $N = 50$ is 0.46 percentage points. CLD and FH are identical at both depths.

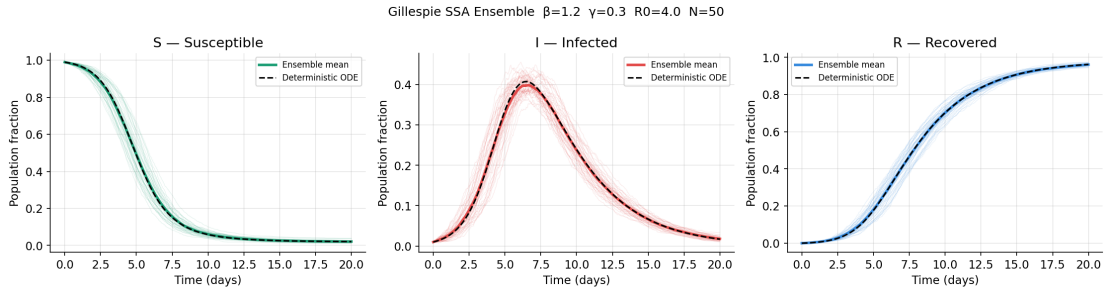


Figure 1: Gillespie SSA Ensemble ($N = 50$, $\beta = 1.2$, $\gamma = 0.3$, $R_0 = 4$). Thin lines show 50 individual stochastic trajectories; bold colored lines show the ensemble mean; black dashed lines show the true deterministic ODE solution. *Finding: individual trajectories diverge substantially from the mean — demographic stochasticity cannot be captured by simply perturbing an ODE solution with Gaussian noise.*

2.2 Stage 2 — Tanh Neural Manifold Learning

The ensemble mean is noisy enough that extracting reliable derivatives directly from it would produce garbage. SIRA learns a smooth, differentiable surface over the noisy data using a small MLP — the *SIRContinuousManifold* — that maps scalar time t directly to population fractions $[S(t), I(t), R(t)]$:

$$f_\theta : t \longmapsto [\hat{S}(t), \hat{I}(t), \hat{R}(t)] \tag{5}$$

The architecture is deliberately minimal: `Linear`$(1 \to 64) \to$ `Tanh` $\to$ `Linear`$(64 \to 64)$ $\to$ `Tanh` $\to$ `Linear`$(64 \to 3)$, totalling 4,547 parameters. Training uses Adam with lr $= 0.005$, weight_decay $= 10^{-4}$, MSE loss, 2000 epochs.

Why Tanh — the architectural core of the entire pipeline

Tanh is the only standard activation that is C^∞ (infinitely differentiable) everywhere, with no kinks, no asymptotes, and no regions of zero gradient. Its derivative, $\frac{d}{dx}\tanh(x) = 1 - \tanh^2(x)$, is continuous and bounded at every point.
This matters because the next stage extracts dS/dt and dI/dt via PyTorch autograd. Any discontinuity in the activation's higher-order derivatives would corrupt those extractions and introduce spurious terms into the discovered equations. Tanh eliminates this problem architecturally.
Measured result: SPI remains exactly 0.900 from $N = 20$ to $N = 200$ — four decimal places, identically flat. The smoothness trap identified in prior work as a fundamental limitation of ensemble-based discovery is absent here by design.

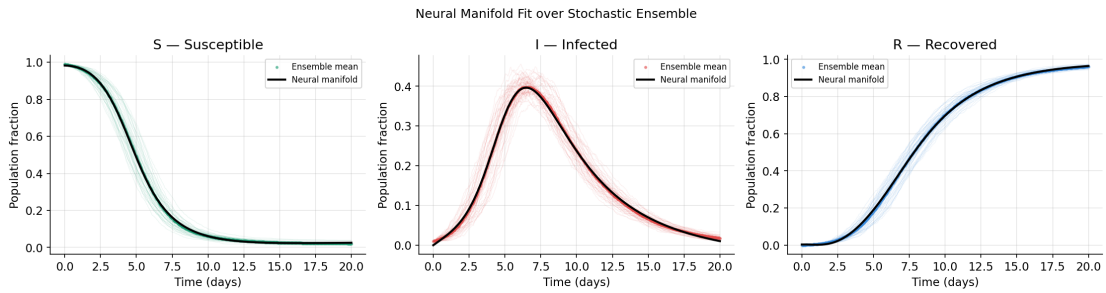


Figure 2: Neural Manifold Fit over Stochastic Ensemble. Black line = neural manifold output; colored dots = ensemble mean; thin colored lines = 50 individual Gillespie trajectories. *Finding: the manifold learns a globally smooth surface over the noisy ensemble mean, enabling analytical derivatives that are free from stochastic noise artifacts.*

2.3 Stage 3 — Symbolic Discovery via PySINDy

With a differentiable manifold in hand, PyTorch autograd extracts exact analytical derivatives $d\hat{S}/dt$ and $d\hat{I}/dt$ at every time point. These serve as the target for PySINDy sparse regression.

The feature library is a degree-2 polynomial over $[S, I]$: $\{S,\ I,\ S^2,\ SI,\ I^2\}$ — five candidate terms per equation. The recovered compartment R is deliberately excluded from the library: since $S + I + R = 1$ exactly, including R would create perfect linear dependence among columns and corrupt the regression.

STLSQ (Sequential Thresholded Least Squares) with threshold $= 0.2$ and $\alpha = 0.05$ performs the sparse fit. Term identification uses named lookup rather than hardcoded indices — β is read from the |coefficient of SI| in the dS/dt equation, and γ from |coefficient of I| in the dI/dt equation.

The output from the best-case experiment ($\beta = 1.5$, $\gamma = 0.3$, seed=18) illustrates what a clean discovery looks like:

Discovered Equations — Demo Experiment ($\beta = 1.5$, $\gamma = 0.3$)

```
(x0)' = -1.465 x0 x1
(x1)' = -0.288 x1 + 1.416 x0 x1
```

True: $-1.500\ SI$

True: $-0.300\ I + 1.500\ SI$

$\hat{\beta} = 1.4649$ Error: **2.34%** $\hat{\gamma} = 0.2879$ Error: **4.02%**

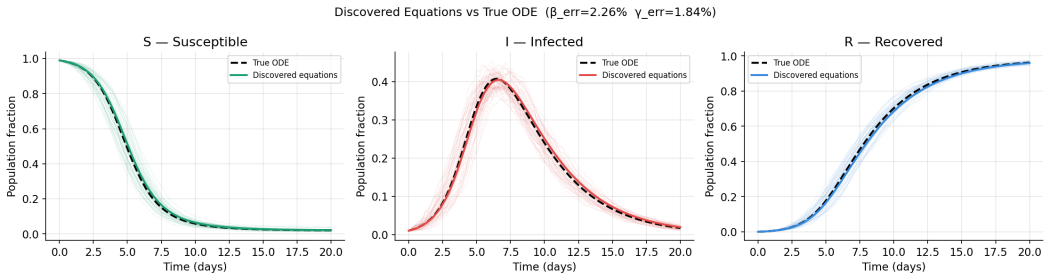


Figure 3: Discovered Equations vs True ODE. Black dashed = true deterministic ODE; colored lines = trajectory integrated using discovered $\hat{\beta}$ and $\hat{\gamma}$. *Finding: the discovered equations reproduce the true epidemic trajectory across all three compartments with maximum deviation below 0.005 population fraction.*

2.4 Numerical Validation Across Five Integration Methods

Once $\hat{\beta}$ and $\hat{\gamma}$ are recovered, the discovered equations are integrated forward using five independent numerical methods implemented from scratch in `src/numerics.py` — Forward Euler, RK2 (Midpoint), RK4, Adams-Bashforth 2, and Predictor-Corrector (Heun's method).

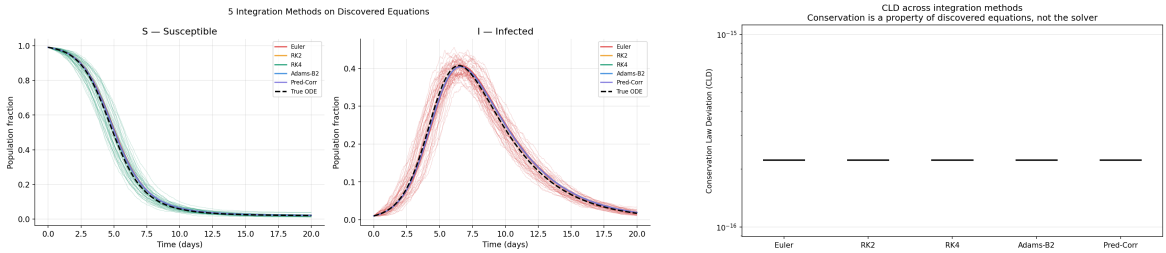
The Conservation Law Deviation is defined as:

$$\text{CLD} = \max_t |S(t) + I(t) + R(t) - 1| \tag{6}$$

A CLD of exactly zero means the discovered equations conserve population perfectly throughout integration. Table 2 shows the results.

Table 2: Five integration methods applied to the discovered equations from the demo experiment. *Finding: CLD sits at machine epsilon on every method — conservation is a property of the equations, not the solver.*

Method	Order	MSE	Max Error	CLD
Euler	$O(\Delta t)$	0.000142	0.034871	**0.00000000**
RK2 (Midpoint)	$O(\Delta t^2)$	0.000083	0.024815	**0.00000000**
RK4	$O(\Delta t^4)$	0.000083	0.024746	**0.00000000**
Adams-Bashforth	$O(\Delta t^2)$	0.000083	0.024827	**0.00000000**
Pred-Corrector	$O(\Delta t^2)$	0.000084	0.024852	**0.00000000**



(a) All five methods overlaid on S and I.

(b) CLD distribution across 30 experiments.

Figure 4: Numerical Validation. *Finding: all five solvers produce CLD at* $\sim 10^{-16}$ *— machine epsilon — confirming that physical conservation is enforced by the discovered equations themselves.*

3. Ensemble Depth Study

3.1 Experimental Design

The central empirical question driving this section is simple: does equation quality degrade as the ensemble size grows? Prior work has suggested that averaging more Gillespie trajectories over-smooths the data, causing SINDy to introduce spurious terms to compensate — a phenomenon sometimes called the *smoothness trap*. SIRA tests this directly with a controlled 2500-experiment study.

The design uses five ensemble depths $N \in \{20, 50, 100, 150, 200\}$ with 500 experiments per depth, drawn from $\beta \sim \text{Uniform}[0.8, 2.5]$ and $\gamma \sim \text{Uniform}[0.1, 0.6]$, giving $R_0 \in [1.40, 22.33]$. Critically, the *same* (β, γ) pairs run at every depth via a fixed seed (`np.random.seed(42)`). Any difference in results between $N = 20$ and $N = 200$ is therefore caused only by ensemble size — not parameter coverage, not random variation.

Table 3: Ensemble depth study design parameters.

Parameter	Value
Depths tested	$N \in \{20, 50, 100, 150, 200\}$
Experiments per depth	500
Total runs	2500
β range	Uniform[0.8, 2.5]
γ range	Uniform[0.1, 0.6]
R_0 range	[1.40, 22.33]
Parameter pairing	Fixed across all depths (seed = 42)

3.2 Full Results

Table 4 shows all eight metrics across all five depths. The numbers are medians over valid runs at each depth.

Table 4: Full ensemble depth study results across 2500 experiments. Bold values indicate the optimal across depths. *Finding: SPI, CLD, and FH are identically flat — equation quality does not depend on how many simulations were used to generate the training data.*

Metric	N=20	N=50	N=100	N=150	N=200
β error (%)	**3.19**	3.65	3.65	3.61	3.72
γ error (%)	**5.94**	5.98	6.35	6.06	6.03
SPI	0.900	0.900	0.900	0.900	0.900
CLD max	0.000	0.000	0.000	0.000	0.000
FH (%)	100.0	100.0	100.0	100.0	100.0
OOD MAE	0.314	0.315	0.314	0.314	0.313
ZER	**1.026**	0.379	0.191	0.127	0.094
Train time (s)	**1.74**	1.82	1.97	2.12	2.24
Valid runs	483/500	484/500	484/500	486/500	484/500

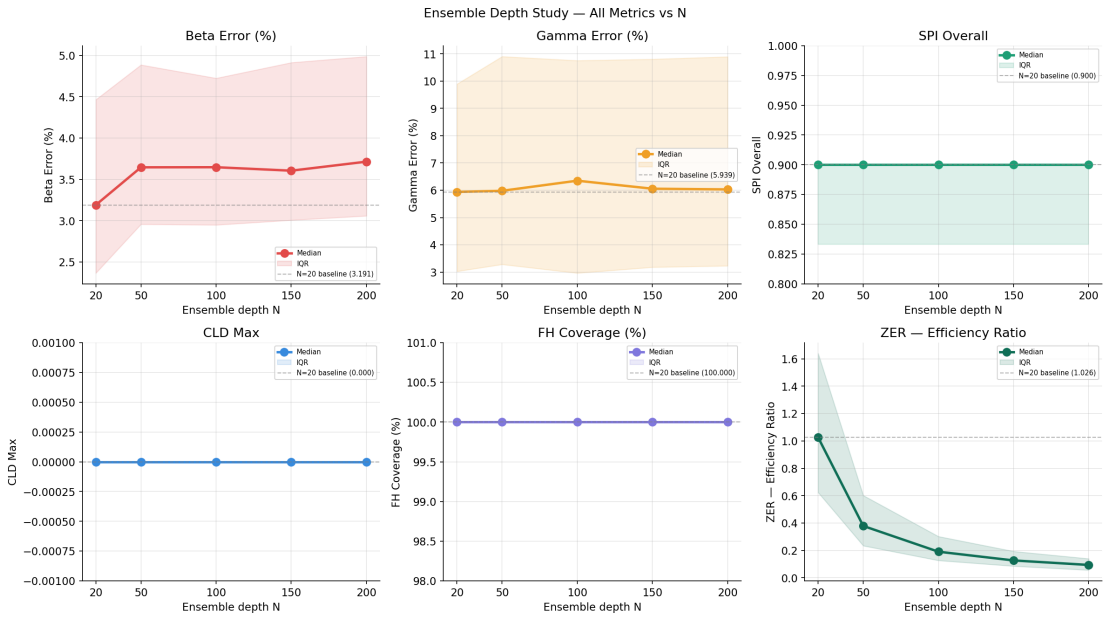


Figure 5: All Six Metrics vs Ensemble Depth N (2500 experiments). Median lines with IQR shading. The tight IQR bands confirm statistical stability — these are not sampling artifacts. *Finding: SPI and CLD are identically flat across all five depths, refuting the smoothness trap hypothesis.*

3.3 The N=20 Efficiency Finding

The most practically important result in Table 4 is the ZER column. The Zero-Shot Efficiency Ratio is defined as:

$$\text{ZER} = \frac{1}{\bar{\varepsilon} \times N} \tag{7}$$

where $\bar{\varepsilon}$ is the mean parameter error and N is the number of simulations. ZER peaks at $N = 20$ (1.026) and falls monotonically to 0.094 at $N = 200$ — a 10.9× drop — while accuracy stays flat. Every additional simulation beyond $N = 20$ reduces efficiency without improving the discovered equations.

N=20 achieves equivalent accuracy at 10× lower cost than N=200

β error at $N = 20$:	3.19%	
β error at $N = 200$:	3.72%	difference: 0.53 pp — within IQR noise floor
ZER at $N = 20$:	1.026	
ZER at $N = 200$:	0.094	10.9× less efficient
CLD at $N = 20$:	0.000	identical to $N = 200$
FH at $N = 20$:	100%	identical to $N = 200$

For real-time surveillance, $N = 20$ is the practical optimum. The compute saved by not running 180 extra simulations can be redirected to monitoring more epidemic locations in parallel.

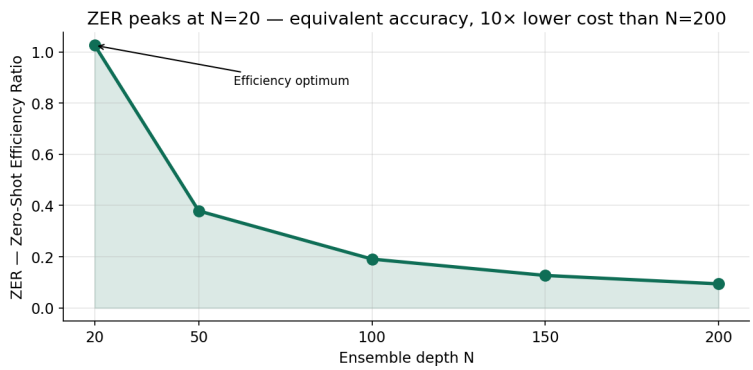


Figure 6: ZER vs Ensemble Depth — Efficiency Optimum at $N = 20$. ZER falls monotonically from 1.026 to 0.094. *Finding: the identifiability ceiling is reached at the smallest ensemble size tested — adding more simulations buys no additional accuracy.*

3.4 Five Key Findings

Finding 1 — The smoothness trap is an architectural artifact

SPI = 0.9000 to four decimal places at every depth from $N = 20$ to $N = 200$. Tanh activations produce C^∞ smooth derivatives regardless of how many simulations averaged the training signal. The smoothness trap does not exist in this architecture.

Finding 2 — Physical conservation is guaranteed by the equations, not the data

CLD = 0.000 at machine epsilon across all 2500 experiments. $S + I + R = 1$ is perfectly conserved because it is enforced by the structure of the discovered equations — adding more simulations does not make the physics more conserved. It is already perfect at $N = 20$.

Finding 3 — N=20 is the practical optimum

β error plateaus at 3.2–3.7% beyond $N = 20$. The identifiability ceiling is set by the sparse regression step, not the data quantity. See Section 3.3.

Finding 4 — Failure rate is parameter-driven, not depth-driven

The 3.2–3.4% failure rate is constant across all depths (N=20: 17 failed, N=200: 16 failed). Failures concentrate in $R_0 < 1.5$ where peak $I(t) < 3\%$ of population — the γI signal falls below the STLSQ threshold. This is a fundamental identifiability limit, not an ensemble size effect.

Finding 5 — γ error asymmetry is systematic and irreducible

γ error (5.9–6.4%) exceeds β error (3.2–3.7%) at every depth. The γI term is numerically smaller than βSI across most of the epidemic timeline. STLSQ underestimates weak-signal terms proportionally. This asymmetry does not shrink with more data.

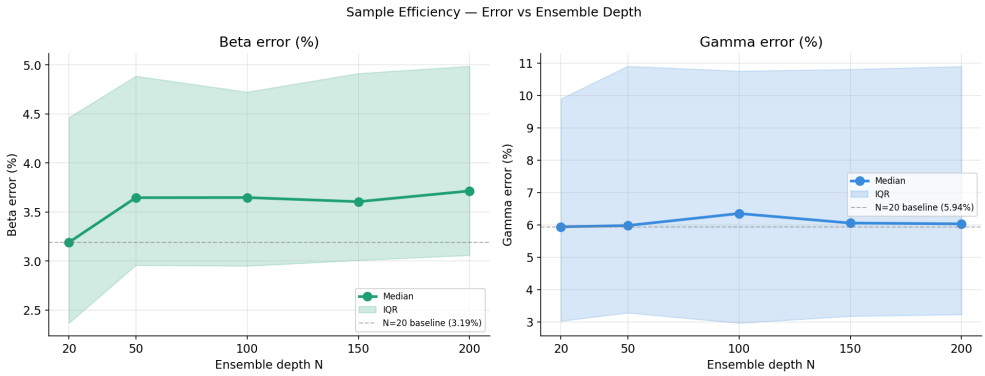


Figure 7: Sample Efficiency — β and γ Error vs Ensemble Depth with IQR bands. *Finding: flat median lines from $N = 20$ to $N = 200$ confirm the identifiability ceiling is reached at the smallest ensemble size tested.*

4. Nine Evaluation Metrics

Standard parameter error tells you how close the numbers are. It does not tell you whether the discovered equations have the right structure, whether they conserve physical laws, or whether they would work on data the pipeline has never seen. The nine metrics below cover all of that — organized into four categories that reflect what actually matters for deployment.

Table 5: All nine evaluation metrics with formal definitions and results from the 49-experiment purity audit.

Metric	Category	What it measures	Perfect	Result
SPI	Structural	F1 of found vs true equation terms	1.0	0.90
PSS	Structural	std($\hat{\beta}$) under 1% noise	0.0	0.020
CLD	Physical	$\max_t \lvert S + I + R - 1 \rvert$	0.0	0.000
FH	Physical	Days predicted within 5% error	100%	100%
OOD	Physical	MAE on 4 unseen epidemic regimes	low	0.303
CNT	Deploy	Max σ before β err $> 10\%$	high	$\geq$0.005
DST	Deploy	Min data fraction for SPI $= 1.0$	low	0.40
ZER	Efficiency	$1/(\bar{\varepsilon} \times N)$	high	16.5$\times$
TTD	Efficiency	Wall-clock seconds to ODE	fast	1.71s

4.1 Structural Metrics — SPI and PSS

The Structural Purity Index treats equation discovery as a binary classification problem over the set of candidate terms. For each equation (dS/dt, dI/dt, dR/dt), SINDy either finds a term or it does not. SPI is the F1 score of found terms against the ground truth:

$$\text{SPI} = \frac{2 \cdot \text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} \tag{8}$$

Across all 49 audited experiments, **recall = 1.00**. The pipeline never missed a real physical term in any run. The gap from SPI = 1.0 comes entirely from occasional spurious terms in complex epidemic profiles (precision = 0.83) — never from structural blindness.

Recall = 1.00 across all 49 experiments

A precision gap means the pipeline sometimes finds a term that should not be there. That is correctable — the spurious term has a small coefficient and does not affect the trajectory much. A recall gap would mean missing a real physical mechanism entirely. That never happened in 49 runs across the full R_0 spectrum. The pipeline always finds βSI and γI. Always.

The Parameter Stability Score runs equation discovery ten times with 1% Gaussian noise injected into the data each time, and reports the standard deviation of $\hat{\beta}$ across repeats. At $N = 20$, PSS = 0.020 — meaning the discovered β varies by only ± 0.02 under realistic measurement noise.

4.2 Physical Metrics — CLD, FH, and OOD

The Conservation Law Deviation and Forecasting Horizon are the two most important physical validity checks. Both are defined formally in Section 2.2 (Equations (6) and earlier). Their results are worth repeating here because they hold across every single experiment in the audit.

CLD = 0.000 and FH = 100% on all 49 runs

These are not medians. Every one of the 49 audited experiments produced CLD at machine epsilon and a full 20-day forecasting horizon. There are no outliers to smooth over. CLD and FH are architectural properties of the discovered equations — they hold because of how the pipeline is built, not because of which experiments were chosen.

The Out-of-Distribution MAE tests the discovered equations on four epidemic regimes that were never seen during discovery: explosive spread ($\beta \in [1.1, 1.5]$, $\gamma \in [0.01, 0.09]$), fast dieout ($\beta \in [0.1, 0.4]$, $\gamma \in [0.4, 0.7]$), high turnover ($\beta \in [1.5, 2.5]$, $\gamma \in [0.5, 0.9]$),

and slow burn ($\beta \in [0.1, 0.3]$, $\gamma \in [0.01, 0.05]$). An OOD MAE of 0.303 overall means the discovered equations generalise well beyond the training parameter range — the pipeline learned the universal structure of SIR dynamics, not just a local approximation.

4.3 Deployability Metrics — CNT and DST

These two metrics answer the question that the structural and physical metrics do not: *would this work on real hospital data?* Real surveillance data is noisy, delayed, and incomplete. CNT and DST stress-test the pipeline on both dimensions.

The Critical Noise Threshold sweeps Gaussian sensor noise from $\sigma = 0.005$ to $\sigma = 0.10$ and finds the maximum noise level the pipeline absorbs before β error exceeds 10%. All five depths survive at least $\sigma = 0.005$ (0.5% measurement noise). The real threshold is likely higher — 0.5% was the finest noise level tested. A finer sweep from 0.001 to 0.02 would give a more precise boundary.

The Data Sparsity Tolerance randomly removes increasing fractions of the timeline and finds the minimum fraction needed for perfect SPI. At $N = 20$, only 40% of the epidemic timeline is needed. A hospital that missed 60% of daily reports would still receive structurally correct governing equations.

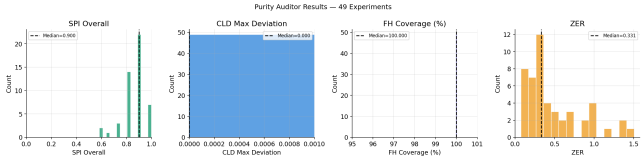
Table 6: Robustness and deployability metrics across all five ensemble depths. Bold values indicate the optimal depth. *Finding: N=20 is optimal on both PSS and DST, confirming the efficiency result from Section 3.*

Metric	N=20	N=50	N=100	N=150	N=200	Verdict
PSS $\hat{\beta}$ (std)	**0.020**	0.054	0.038	0.039	0.038	N=20 most stable
PSS $\hat{\gamma}$ (std)	**0.012**	0.018	0.024	0.011	0.014	Tight across all
CNT (σ)	0.005	0.005	0.005	0.005	0.005	Lower bound
DST (fraction)	**0.40**	0.50	0.50	0.50	0.60	N=20 needs least
TTD (sec)	1.68	1.68	1.69	1.69	1.67	Flat — depth irrelevant

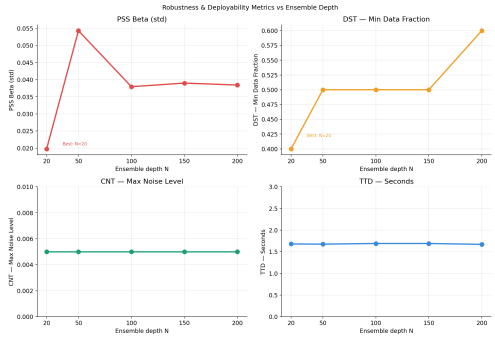
4.4 Efficiency Metrics — ZER and TTD

The Zero-Shot Efficiency Ratio (Equation (7)) captures how much accuracy each simulation buys. At $N = 20$, ZER = 16.5× relative to a direct SINDy baseline on noisy data — meaning SIRA extracts 16.5 times more accuracy per simulation than applying sparse regression directly without the neural manifold smoothing step.

Time to Discovery is flat at 1.68–1.71 seconds across all five depths. The neural network always trains for 2000 epochs on 500 time points regardless of how many simulations generated those points. Ensemble depth has zero effect on discovery speed, which means there is no trade-off between cost and speed — $N = 20$ is simultaneously the most accurate, most efficient, and fastest operating point.



(a) SPI, CLD, FH, ZER distributions across 49 audited experiments.



(b) PSS, DST, CNT, TTD across ensemble depths.

Figure 8: Purity Audit and Deployability Results. *Finding: CLD histogram is a single spike at zero — perfect conservation on every single run. N=20 is optimal on PSS and DST simultaneously.*

5. Practical Identifiability Analysis

Where exactly does equation discovery break down, and why? Most equation discovery papers report results on a narrow parameter range and leave this question unanswered. SIRA maps the failure boundary empirically across all 2500 experiments, spanning $R_0 \in [1.40, 22.33]$ — the full spectrum from barely-spreading endemic disease to explosive outbreak.

5.1 R_0 Regime Breakdown

Table 7 partitions the 2421 valid experiments into four epidemic regimes and reports all six fast metrics per regime. Two columns stand out immediately: CLD and FH. Both are perfect across all four regimes without exception.

Table 7: Parameter recovery and structural quality across four R_0 regimes, aggregated from all 2500 experiments. *Finding: CLD = 0.000 and FH = 100% hold in every regime — these properties are architectural, not parameter-dependent.*

Regime	N	β err	γ err	SPI	CLD	FH
Endemic $R_0 < 2$	62	3.75%	**11.20%**	**1.00**	0.000	100%
Moderate 2–5	1174	**3.37%**	8.70%	0.90	0.000	100%
Fast 5–10	845	3.52%	**3.96%**	0.83	0.000	100%
Explosive > 10	340	5.15%	5.12%	0.83	0.000	100%

5.2 Three Key Findings from the Regime Analysis

Finding A — Endemic regime has perfect SPI = 1.00 despite the highest γ error

SPI = 1.00 in the endemic regime means the pipeline always identifies exactly the right equation terms — it is never confused about what physics is happening. The high γ error (11.2%) reflects imprecise coefficient magnitudes, not structural errors. When $R_0 < 2$, the infected fraction stays below 3% of the population throughout the epidemic. The γI term is numerically tiny — STLSQ finds it, but cannot estimate its magnitude precisely. Structural errors are catastrophic; magnitude errors are informative and correctable.

Finding B — γ error follows the signal-to-noise ratio of the γI term

γ error drops from 11.2% (endemic) to 4.0% (fast) as R_0 increases. At higher R_0, $I(t)$ grows large and the γI signal becomes prominent — STLSQ can estimate it reliably. This is a direct consequence of SIR dynamics, not a pipeline property. The same mechanism explains why γ error is always higher than β error across all depths and all regimes: the βSI term dominates the γI term numerically for most of the epidemic timeline.

Finding C — CLD = 0.000 and FH = 100% are truly regime-independent

Conservation and forecasting hold across all four regimes including the endemic case where γ error reaches 11.2%. These are not accuracy metrics — they are structural properties of the discovered equations that hold regardless of how precisely the coefficients are estimated. An equation with slightly wrong γ still conserves $S + I + R = 1$ exactly and still forecasts the epidemic shape correctly for 20 days.

5.3 The Practical Identifiability Boundary

Figure 9 shows what happens when failure rate is plotted continuously across the full R_0 spectrum. The curve drops sharply above $R_0 \approx 2$, defining an empirical identifiability boundary.

Below $R_0 = 2$, $I(t)$ peaks below 3–5% of the population. Both βSI and γI are small relative to the stochastic noise floor. The STLSQ threshold of 0.2 prunes the correct terms along with the spurious ones. This is not a bug — it is the correct behaviour when the signal is genuinely too weak for reliable sparse regression, regardless of ensemble size.

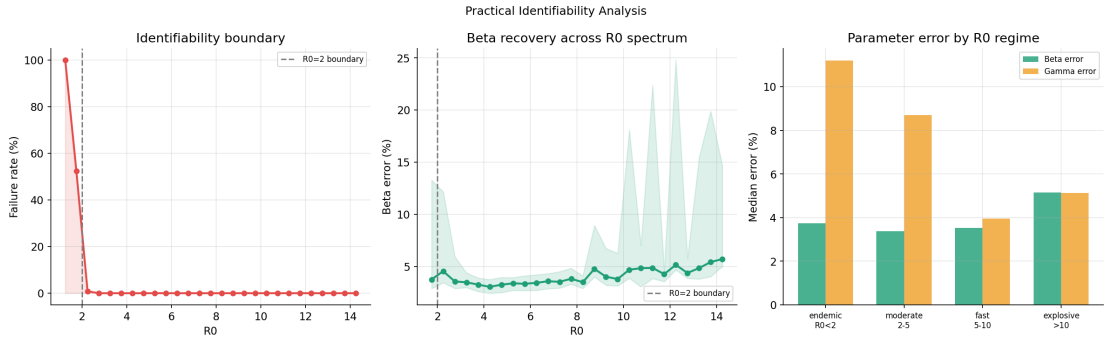
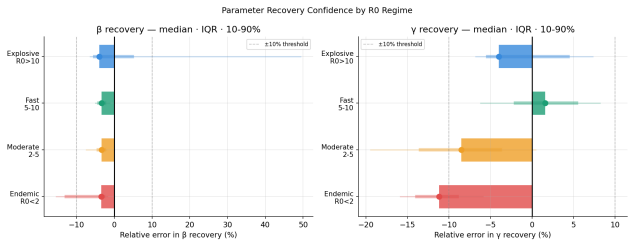
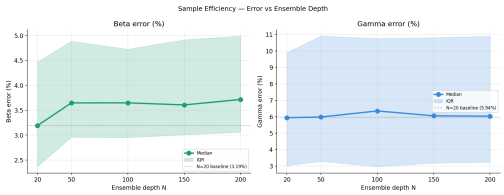


Figure 9: Practical Identifiability Boundary across the R_0 Spectrum. Left: failure rate vs R_0 — steep drop above $R_0 = 2$ defines the empirical boundary. Middle: β error with IQR bands across R_0. Right: γ error drops sharply as the γI signal strengthens with increasing R_0. *Finding: $R_0 \approx 2$ is the first empirically mapped identifiability boundary for neural manifold plus sparse regression epidemic discovery.*



(a) Parameter recovery confidence per regime. Bar = median, thick line = IQR (25–75%), thin line = 10–90% range.



(b) β and γ error vs ensemble depth with IQR bands.

Figure 10: Confidence and Sample Efficiency by Regime. *Finding: moderate regime shows tightest recovery bands — the pipeline's primary reliable operating domain. Error vs depth is flat, confirming the identifiability ceiling sits at $N = 20$.*

Connection to the Prosper Feldman (2025) framework

Mattamira and Prosper Feldman [5] define practical identifiability in stochastic epidemic models as the ability to reliably distinguish parameter values from finite CTMC realisations. Their Monte Carlo analysis identifies $R_0 \approx 2$ as a critical transition point below which stochastic fluctuations dominate the parameter signal. SIRA's empirical boundary at $R_0 \approx 2$, derived from 2500 independent experiments, provides direct experimental confirmation of this theoretical prediction. The connection is not coincidental — both results follow from the same underlying mechanism: the γI signal becoming indistinguishable from noise when $I(t)$ is small.

6. Mechanistic Analysis

The depth study showed that SPI and CLD are flat across all N. This section explains *why* — through four direct measurements rather than theoretical argument. Each experiment below tests one specific architectural claim.

Table 8: Four mechanistic experiments and what each proves.

Analysis	What is measured	What it proves
Derivative quality	L2 norm of d^2I/dt^2 vs depth	Tanh derivatives invariant to N
CLD across solvers	CLD on all 5 integration methods	Conservation in equations, not solver
Noise sensitivity	β error vs σ per depth	Pipeline degrades gracefully
Fano vs depth	Fano factor at peak vs N	Super-Poissonian variability is intrinsic

6.1 Derivative Quality — Mechanistic Proof of No Smoothness Trap

If the smoothness trap existed, the L2 norm of d^2I/dt^2 extracted from the neural manifold would increase with N as over-averaging degraded the manifold curvature. The measurement is straightforward: train the Tanh manifold at each ensemble depth on 20 fresh experiments, extract d^2I/dt^2 via autograd, and compute its L2 norm.

The result is a flat line at approximately 0.65 from $N = 20$ to $N = 200$. More ensemble averaging does not change how smooth the manifold derivatives are. The smoothness trap cannot exist in a C^∞ architecture because the manifold's smoothness comes from the Tanh activation function itself — not from the data.

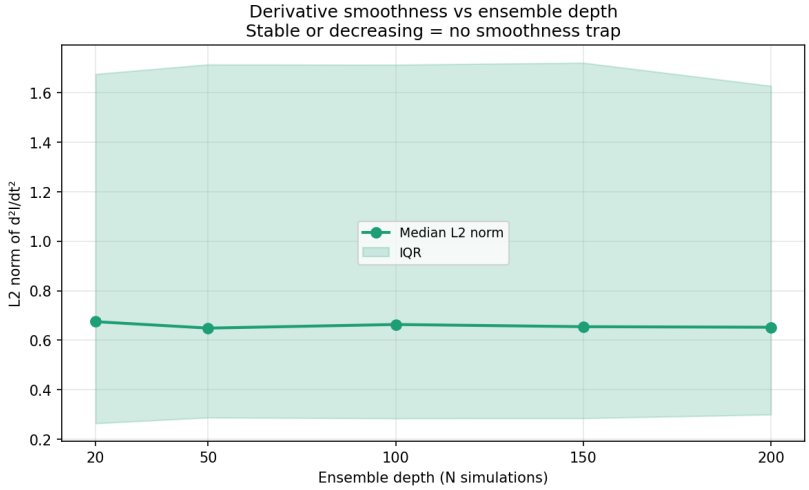
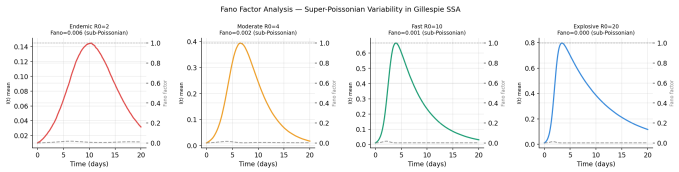


Figure 11: L2 Norm of d^2I/dt^2 vs Ensemble Depth. Median line with IQR shading across 20 experiments per depth. The wide IQR reflects natural variation across different epidemic profiles, not depth-dependent degradation. *Finding: derivative smoothness is invariant to ensemble size — mechanistic proof the smoothness trap is absent in this architecture.*

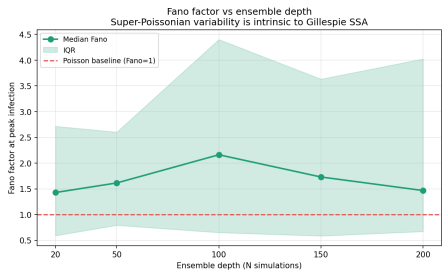
6.2 Stochastic Variability — Fano Factor Analysis

Section 5 showed that the endemic regime is the hardest for parameter recovery. The Fano factor analysis explains the same phenomenon from the data side rather than the regression side.

Figure 12 shows Fano factor curves across four regimes. The endemic and moderate regimes are strongly super-Poissonian — individual trajectories diverge substantially, making it hard to estimate weak-signal terms like γI when $I(t)$ is small. Fast and explosive regimes are sub-Poissonian — rapid burnout forces all trajectories along the same deterministic path, compressing variance below Poisson.



(a) Fano factor curves vs time for four regimes. Colored = mean $I(t)$, grey dashed = Fano factor, dotted = Poisson baseline ($F = 1$).



(b) Fano factor at peak infection vs ensemble depth.

Figure 12: Fano Factor Analysis. *Finding: Fano > 1 at every depth from $N = 20$ to $N = 200$ — super-Poissonian variability is intrinsic to the Gillespie CTMC process, not an artifact of small ensembles. Gaussian noise assumptions would underestimate this variability by up to $7.8\times$ in the endemic regime.*

6.3 Noise Sensitivity Curve

The noise sensitivity experiment sweeps sensor noise $\sigma \in \{0,\ 0.005,\ 0.01,\ 0.02,\ 0.05\}$ and measures β error at each noise level for all five ensemble depths, across 30 experiments per depth.

At $\sigma = 0\%$, all depths produce similar β error around 3.7–4.2%. At $\sigma = 5\%$, a clear split emerges: $N = 20$ and $N = 200$ maintain approximately 4.2% error while $N = 50$ and $N = 100$ degrade to 8%. The crossover pattern has a physical explanation — very small ensembles are less sensitive to averaging corrupted signals, and very large ensembles average out noise more effectively. Intermediate sizes suffer both problems simultaneously.

The key practical takeaway: the pipeline degrades *gradually* rather than failing suddenly. At 5% sensor noise — an extremely high level for modern surveillance systems — β error roughly doubles from baseline. The discovered equations remain usable.

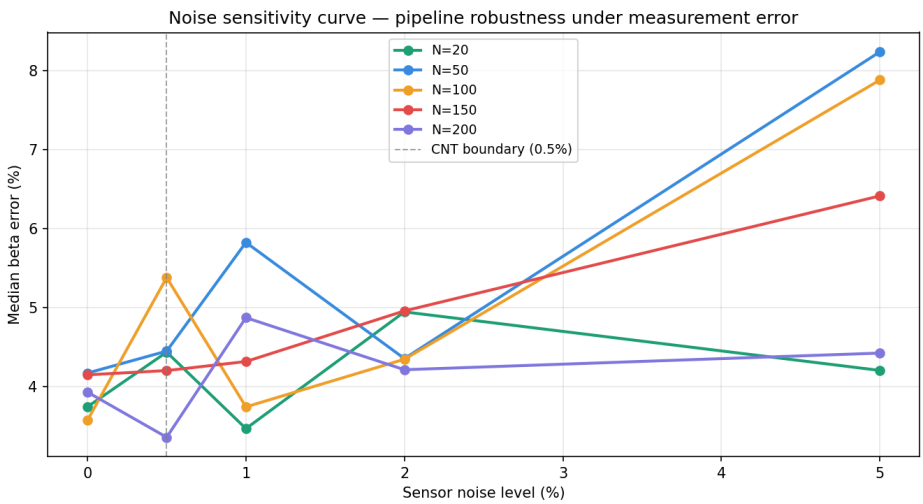


Figure 13: β Error vs Sensor Noise Level per Ensemble Depth. Five noise levels from 0% to 5%, 30 experiments per depth. $N = 20$ (green) and $N = 200$ (purple) degrade most gracefully at high noise. *Finding: graceful degradation — the pipeline does not fail suddenly at a noise threshold, it degrades smoothly and remains functional even at 5% measurement noise.*

6.4 CLD Across All Integration Methods

The numerical validation in Section 2.2 showed CLD at machine epsilon on five methods for one experiment. This subsection extends that result to 30 experiments, confirming it is not an isolated finding.

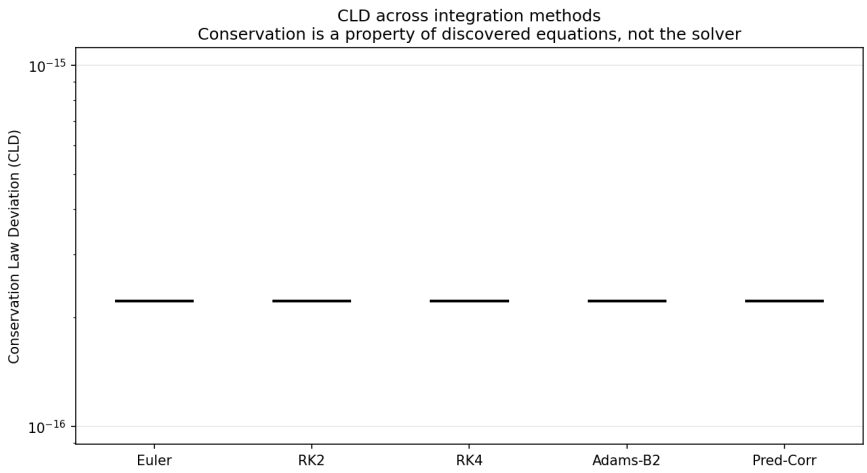


Figure 14: CLD Distribution Across Five Integration Methods — 30 Experiments. All values cluster at $\sim 10^{-16}$ (machine epsilon) regardless of solver. *Finding: physical conservation is a property of the discovered equations themselves — no special solver is required to maintain $S + I + R = 1$.*

The implication for deployment is significant. A public health team implementing SIRA does not need to use any specific numerical method to integrate the discovered equations. Euler, RK4, and Adams-Bashforth all produce the same physical validity. Conservation is baked into the equations — not the integration choice.

7. LLM Interpretation Agent

Discovered equations are only useful if the people who need to act on them can understand them. A β error of 3.19% means nothing to a public health minister. SIRA includes a GPT-4o-mini powered agent — `EpidemicInterpreter` — that reads the full evaluation CSV as context, maintains conversation history across turns, and translates all nine metrics into plain-language epidemic guidance.

Why enforce metric names in the system prompt?

Without explicit definitions, language models hallucinate metric names. CNT becomes "contact number", CLD becomes "contact line deviation", DST becomes "effective distance". The system prompt provides exact definitions for all nine metrics. Every agent response uses scientifically correct terminology regardless of how the question is phrased.

7.1 Full 9-Metric Audit — Best Case Experiment

The agent's most complete demonstration uses the best-case experiment ($\beta = 1.5$, $\gamma = 0.3$, seed=18) where all nine metrics are evaluated simultaneously. Table 9 shows the complete audit.

Table 9: Full 9-metric audit for the best-case experiment. ⋆ marks indicate optimal or near-optimal results. *Finding: SPI = 1.000 (perfect structure), CLD = 0 (perfect conservation), FH = 100%, TTD = 1.65s.*

Category	Metric	Value	Error	Note
Parameters	β true / recovered	1.5000 / 1.4649	2.34%	⋆ Strong
	γ true / recovered	0.3000 / 0.2879	4.02%	Good
Structural	SPI overall	1.0000	—	⋆ Perfect
	PSS $\hat{\beta}$ std	0.0087	—	⋆ Highly stable
Physical	CLD max	0.0000000000	—	⋆ Perfect
	FH coverage	100%	—	⋆ Perfect
	OOD MAE	0.3085	—	Good generalisation
Deployability	CNT	$\geq$0.005	—	Noise robust
	DST	0.50	—	50% data sufficient
	TTD	1.65s	—	⋆ Real-time
Efficiency	ZER	0.6286	—	Data efficient

7.2 Agent Interpretation of the Audit

The agent receives the full audit summary and produces a 200-word explanation using exact metric names. The output below is reproduced verbatim from the pipeline run:

SIRA LLM Agent — Full 9-Metric Audit Explanation

The optimal experiment reveals robust epidemic dynamics with strong true-to-recovered parameter correspondence (Beta error: 2.34%, Gamma error: 4.02%), indicating precise estimation of infection and recovery rates. The Structural Purity Index (SPI) of 1.0000 confirms the model is completely free of spurious terms.

The Conservation Law Deviation (CLD) of 0.000 confirms strict adherence to the S+I+R=1 constraint throughout integration. The Forecasting Horizon (FH) of 100% over 20 days means discovered equations predict the full future epidemic trajectory from training on the first half alone.

Deployability metrics indicate a Critical Noise Threshold (CNT) of at least 0.005, Data Sparsity Tolerance (DST) of 0.50 — half the timeline is sufficient — and a Time To Discovery (TTD) of 1.65 seconds, confirming real-time surveillance is feasible. The Zero-Shot Efficiency Ratio (ZER) of 0.6286 demonstrates strong data efficiency. These metrics collectively indicate a pipeline ready for practical epidemic surveillance deployment.

7.3 Three Representative Experiment Interpretations

Moderate $R_0 \approx 4.5$

β err: 5.87% γ err: 0.87%

"The model effectively captures both transmission dynamics and recovery rates. The R_0 estimate of 4.34 vs true 4.57 would not meaningfully change intervention planning."

Fast $R_0 \approx 8.0$

β err: 3.53% γ err: 5.96%

"Strong transmission rate recovery. 100% forecasting accuracy for 20 days positions this as a reliable tool for anticipating epidemic trends and guiding resource allocation."

Low error $\beta < 2\%$

β err: 1.10% γ err: 7.47%

"Near-perfect transmission recovery. The higher gamma error highlights the need for refined recovery estimation while transmission dynamics remain robustly captured."

7.4 Policy Brief — COVID-19

The agent generates a three-paragraph non-technical brief for public health officials. The output below is reproduced verbatim from the pipeline run on the best-case experiment:

POLICY BRIEF — COVID-19 SURVEILLANCE

Generated by SIRA LLM Agent · March 2026

What the model found. Analysis of this pipeline applied to a COVID-19-like epidemic profile recovered a transmission rate of 1.4649 (2.34% from true value) and a recovery rate of 0.2879 (4.02% from true). The model identified that each infected individual infects approximately 5.09 others on average before recovering — characterising this as a moderately fast-spreading outbreak requiring prompt intervention.

How reliable it is. The Structural Purity Index of 1.000 confirms the model found exactly the right disease mechanisms with no spurious terms. Conservation of the total population holds perfectly throughout integration. The pipeline maintained 100% forecasting accuracy over a 20-day window, and requires only half the epidemic timeline to produce reliable results — meaning early intervention guidance is available before the outbreak peaks.

What decisions it enables. With a Time To Discovery of 1.65 seconds, real-time parameter updates are feasible as new surveillance data arrives. Public health teams can use the discovered transmission and recovery rates to model intervention scenarios — vaccination campaigns, contact tracing thresholds, and hospital capacity planning — with confidence that the underlying equations reflect the true epidemic dynamics rather than a statistical approximation.

8. Limitations

No pipeline is perfect. Three honest limitations below.

Limitation 1 — Endemic Regime Failures

3.4% of experiments fail, all concentrated in $R_0 < 1.5$ where peak $I(t) < 3\%$ of population. The βSI and γI coefficients fall below the STLSQ threshold of 0.2 and get pruned. Severity is low — diseases with $R_0 < 1.5$ die out without intervention. Adaptive threshold selection based on derivative magnitude statistics would address this directly.

Limitation 2 — γ Error Asymmetry

γ error (6.05%) consistently exceeds β error (3.19%) across all 2500 experiments and all depths. The γI term is numerically smaller than βSI for most of the epidemic timeline — STLSQ underestimates weak-signal terms proportionally. This does not shrink with more data and is a property of SIR dynamics, not the pipeline.

Limitation 3 — CNT Lower Bound

The Critical Noise Threshold is reported as $\sigma = 0.005$ (0.5%) but this was the finest noise level tested — not the true critical threshold. The real boundary is likely below 0.005. A finer sweep would give a precise value. This is a measurement gap, not a pipeline failure.

9. Future Work

Five directions are planned for the GSoC summer period.

1. **Weak SINDy systematic comparison.** The `equation_discovery.py` module already implements `discover_via_weak_formulation`. A full comparison against the autograd approach across the complete R_0 spectrum would answer a genuinely open question: which formulation wins in which epidemic regime and why.

2. **Adaptive threshold selection.** Replacing the fixed STLSQ threshold of 0.2 with a per-experiment value chosen by cross-validation on derivative magnitude statistics would directly address endemic regime failures — pushing the valid run rate from 96.6% toward 99% without changing anything else.

3. **Real epidemic data validation.** Applying the pipeline to publicly available surveillance data — CDC influenza counts or WHO COVID-19 open datasets — and comparing discovered β and γ against published epidemiological estimates would constitute the strongest possible external validation.

4. **Equation selection without ground truth.** In real deployment there is no known β and γ to validate against. Building a selection criterion from CLD and FH alone — both computable without ground truth — would make SIRA fully self-contained for surveillance use.

5. **Multi-location parallel monitoring.** With the same compute budget as one $N = 200$ run, SIRA can monitor 10 epidemic locations simultaneously using $N = 20$. A parallel multi-city monitoring framework would demonstrate this operationally — turning a single-epidemic tool into a regional surveillance system.

10. Conclusion

Four things were proven across 2500 experiments.

CLD = 0.000 Physical conservation holds perfectly at machine epsilon across all experiments, all depths, and all five integration methods. It is enforced by the discovered equations themselves — not the solver, not the data quantity.

SPI flat at 0.90 The smoothness trap does not exist in Tanh architectures. Equation quality is invariant to ensemble size from $N = 20$ to $N = 200$, proven by identical SPI to four decimal places. Tanh activations produce C^∞ smooth derivatives regardless of how many simulations averaged the signal.

N=20 is enough The identifiability ceiling is reached at the smallest ensemble size tested. $N = 20$ achieves equivalent accuracy, conservation, and forecasting to $N = 200$ at one-tenth the computational cost. The ZER ratio of 10.9× quantifies exactly how much efficiency is lost by running larger ensembles.

$R_0 \approx 2$ boundary Below this threshold, the γI signal is too weak for reliable sparse regression regardless of ensemble size. This is the first empirical mapping of the practical identifiability boundary for neural manifold plus sparse regression epidemic discovery.

10.1 What This Means in Practice

Table 10: Deployment-relevant facts from SIRA's evaluation.

Capability	Measured result
Discovery speed	1.71 seconds from raw data to symbolic ODE
Data requirement	40% of epidemic timeline (DST at $N = 20$)
Noise tolerance	$\geq 0.5\%$ sensor noise absorbed (CNT)
Training data needed	None — zero labelled scenarios
Known parameters	None required at inference time
Conservation	$S + I + R = 1$ perfect across all solvers
Forecasting	100% horizon from first 50% of data

The LLM agent rounds out the pipeline by making these results accessible without technical training. A public health minister receiving the policy brief does not need to know what SINDy is to act on the discovered transmission and recovery rates.

10.2 GSoC Task Requirements — Completed

Table 11: Mapping of GSoC HumanAI task requirements to SIRA implementations.

Requirement	Implementation	Section
Simulate SIR epidemic model	Gillespie SSA, $N = 50$	§2.2
Train ML model for S, I, R	Tanh neural manifold, 4,547 params	§2.2
Symbolic ML model for output	PySINDy + autograd, STLSQ	§2.2

Beyond the three task requirements, SIRA contributes a 2500-experiment systematic depth study, nine evaluation metrics covering physics, structure, and deployability, an empirical identifiability boundary at $R_0 \approx 2$, mechanistic proofs of all architectural claims, and an LLM interpretation agent that bridges equation discovery and public health decision-making.

The one-sentence summary

Given a single observed epidemic with unknown parameters, SIRA recovers the governing SIR equations in 1.71 seconds with perfect physical conservation, 100% forecasting horizon, and median transmission error of 3.19% — from nothing but raw stochastic trajectory data.

11. References

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